

Urban heat island in Southwark

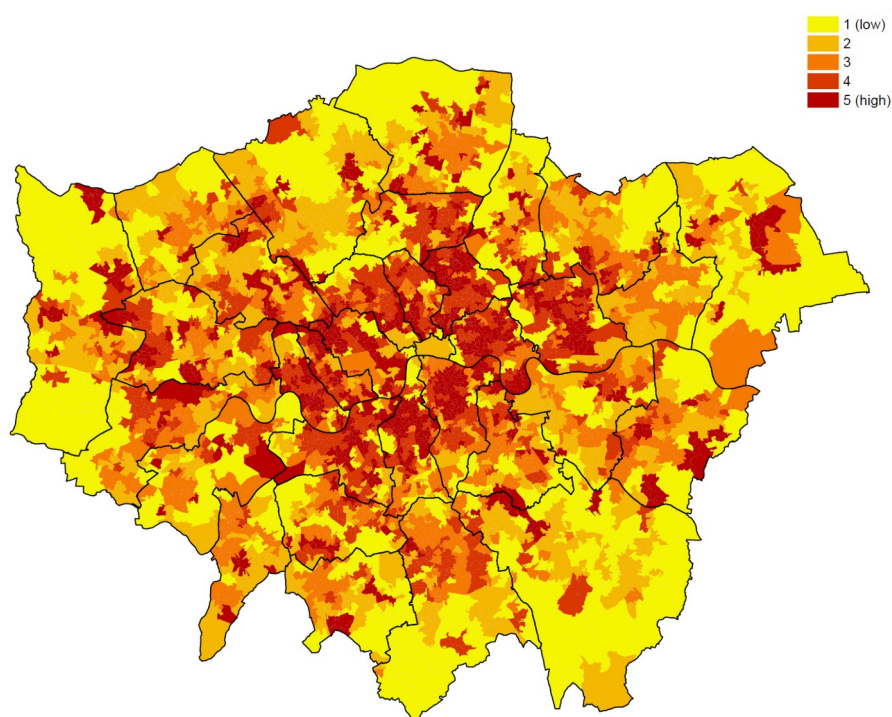
The urban heat island is the extra heat a city holds compared with the countryside around it. Roads, pavements and buildings absorb heat by day and trap it (London Climate Ready Partnership, n.d.). Southwark sits at the centre of London's heat island; central London runs about 4.5°C hotter than its rural surroundings (Arup, 2023).

Evidence

- central London was up to 10°C warmer than rural areas in the 2003 heatwave, and the heat island can add 5 to 6°C to night-time temperatures (London Climate Ready Partnership, n.d.)
- in the July 2022 heatwave the heat island averaged 2.3°C and peaked at 7.2°C, strongest at night and in central London (Simpson, Brousse and Heaviside, 2024)
- it caused an estimated 141 deaths in Greater London over that heatwave, 38% of all heat-related deaths (Simpson, Brousse and Heaviside, 2024)
- 8% of Greater London is a persistent heat island, with mean summer surface temperature of 40.56°C or more over 2013 to 2023; clusters sit in east London, the western edge and south of the Thames (Krenz and Amann, 2025)
- Asian (39.10°C) and Black (39.00°C) residents face higher mean surface temperatures than White residents (38.57°C); children aged 4 and under face the highest of the age groups studied (Krenz and Amann, 2025)
- the least well-off are exposed to surface temperatures 4.38°C higher than the most affluent (90th against 10th percentile) (Krenz and Amann, 2025)
- heat could reduce walking accessibility in affected areas by 6 to 9%, hitting Asian and Black communities hardest (Krenz and Amann, 2025)
- the GLA's heat hazard mapping places Southwark among the central boroughs with a higher heat hazard, due to the heat island (Greater London Authority and Arup, 2024)
- green cover sets the local pattern: Kilburn and South Hampstead (38% vegetation) ran over 7°C hotter than nearby Regent's Park (89%) (Arup, 2023)

UHeatRank map

The GLA's heat hazard map, modelled by Arup with its UHeat tool, ranks every London LSOA from 1 (coolest) to 5 (hottest) by average summer air temperature, June to August 2018 to 2022. Each rank holds a fifth of LSOAs; averages range from 17.0°C to 19.5°C (Greater London Authority and Arup, 2024). Modelled air temperature is closer to what people feel than surface temperature. The ranks average day and night, so they understate heatwave peaks.



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Figure 1. Heat hazard by LSOA, London, June to August 2018 to 2022. Source: Greater London Authority and Arup (2024), Figure 16.

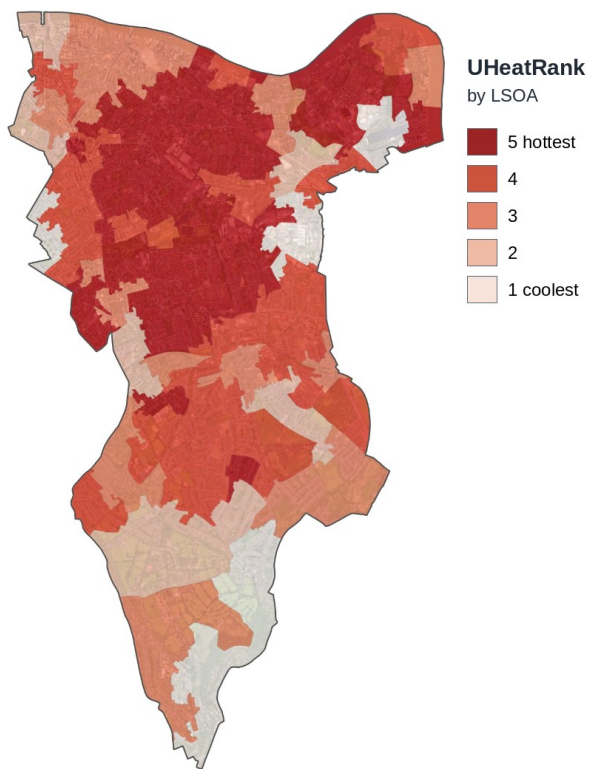


Figure 2. UHeatRank by LSOA, Southwark. Source: Greater London Authority and Arup (2024).

The hottest areas run through the north and centre of the borough; the south and the larger parks rank lowest.

London heat island map

Krenz and Amann (2025) map persistent heat islands across London from Landsat summer imagery, 2013 to 2023, at 30 m resolution. Areas shown have mean surface temperature of 40.56°C or more.

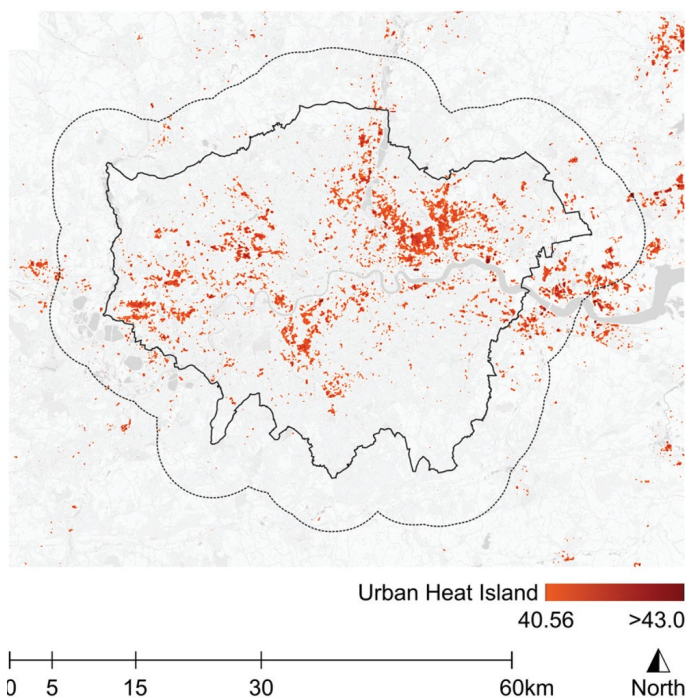


Figure 3. Urban heat island areas, Greater London, 2013 to 2023. Source: Krenz and Amann (2025), Figure 2b, CC BY 4.0.

Land surface temperature

The WRI Cool Cities Lab shows the same pattern at street scale. The map's scale runs from 33°C to 43°C, with the Thames and large green spaces coolest (WRI, 2026). Surface temperature exceeds air temperature, so this shows pattern, not rank.

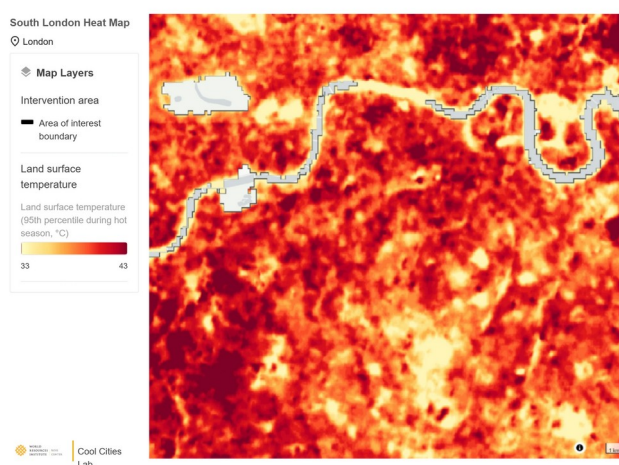


Figure 4. Land surface temperature, 95th percentile hot season (°C). Source: WRI (2026).

Works cited

Arup (2023) Urban Heat Snapshot. London: Arup. Available at: arup.com/insights/urban-heat-snapshot (Accessed: 29 September 2026).

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World Resources Institute (WRI) (2026) Cool Cities Lab. Washington, DC: WRI Ross Center for Sustainable Cities. Available at: coolcities.wri.org (Accessed: 29 September 2026).